上海交通大学学位论文

面向 3C 工业生产线的双臂机器人视觉-语言-动作模型

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A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

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摘 要

现代 3C 电子装配线需要处理小型元器件、高产品多样性和富接触操作，这些任务仍难以用传统固定程序机器人实现自动化。本项目针对这一缺口，面向双臂工业操作开发视觉-语言-动作（Vision-Language-Action, VLA）系统。该系统拟运行于双臂 Franka 平台，并覆盖典型 PC 关键部件任务，具体包括内存条插入、CPU 放置以及电源线连接。

当前实现基于 LeRobot 框架中的 Action Chunking with Transformers（ACT）策略，通过遥操作采集分阶段、任务相关的演示数据，并按目标任务部署对应检查点。现阶段验证聚焦于视觉-动作模仿学习基线；语言条件推理与统一多任务控制将作为后续扩展。最终概念验证目标为：规划成功率高于 90%，任务执行成功率高于 70%，部件检测精度高于 90%，位姿估计误差低于 20 mm，故障检测率高于 90%，恢复率高于 70%。

关键词： 双臂机器人，视觉-语言-动作模型，工业自动化

ABSTRACT

Modern 3C electronics assembly lines deal with small components, high product variety, and contact-rich operations that remain difficult to automate with conventional fixed-program robots. Our project targets this gap by developing a Vision-Language-Action (VLA) system for bimanual industrial manipulation. The system is designed to run on a dual-arm Franka platform and to handle representative PC key-component tasks—specifically RAM insertion, CPU placement, and power-cable connection.

The current implementation uses the Action Chunking with Transformers (ACT) policy in the LeRobot framework. It collects staged, task-specific demonstrations through teleoperation and deploys the checkpoint selected for the requested task. The present validation focuses on a vision–action imitation-learning baseline; language-conditioned inference and unified multi-task control remain later extensions. The final proof-of-concept targets are a planning success rate above 90%, a task execution success rate above 70%, component detection accuracy above 90%, pose estimation error below 20 mm, failure detection rate above 90%, and recovery rate above 70%.

Key words: dual-arm robot, vision-language-action model, industrial automation

Contents

Chapter 1 Introduction

1.1 Background and motivation

The 3C industry—computer, communication, and consumer electronics—is characterized by short product life cycles, frequent model changes, small components, and contact-rich assembly operations. Conventional industrial robots perform well in high-volume production when the workspace, component geometry, and motion sequence remain fixed. However, each product change can require new fixtures, manually programmed trajectories, and repeated commissioning. These requirements increase the cost of automating low-volume and high-mix production, particularly for tasks such as component placement and connector insertion, where success depends on both visual understanding and precise physical interaction.

Learning from demonstration provides a different route to robot programming. Instead of specifying every motion analytically, an operator demonstrates the task and a policy learns to map visual observations and robot proprioceptive states to actions. Action Chunking with Transformers (ACT) and Diffusion Policy have shown that imitation-learning policies can represent temporally coordinated manipulation behaviors from demonstration data[1-2]. Nevertheless, a task-specific visuomotor policy normally has no direct mechanism for interpreting natural-language instructions, and a separate checkpoint is often required for each task.

Vision-Language-Action (VLA) models extend this approach by conditioning robot actions on images, language, and robot state within a unified model. Systems such as RT-2, OpenVLA, RDT-1B, and the π family illustrate the potential of transferring semantic knowledge from large vision-language models to physical control[3-7]. This capability is attractive for flexible manufacturing because one policy can, in principle, distinguish among multiple tasks from an instruction instead of relying on manually selected programs. However, adapting a VLA model to an industrial workcell remains a system-level engineering problem. Reliable deployment requires synchronized multi-view observations, consistent teleoperation data, an action representation compatible with the robot, safe real-time execution, and quantitative evaluation under repeatable physical conditions.

1.2 Project objectives and scope

This project develops and evaluates a language-conditioned manipulation system for a dual-arm Franka Research 3 (FR3) workcell. The hardware platform combines two robot arms, complementary global and wrist-camera views, task-compatible end effectors and fixtures, and a GELLO-based teleoperation interface. Demonstrations are recorded as synchronized camera observations, robot proprioceptive states, and joint-space actions. The resulting pipeline covers the complete process from physical demonstration and dataset preparation to model fine-tuning, inference, and real-robot execution.

The modeling route progresses from visuomotor baselines to a VLA policy. ACT and Diffusion Policy are retained as baselines for evaluating task-specific imitation learning, with ACT providing the more stable initial real-robot deployment. The final system adopts $\pi_{0.5}$, which combines a PaliGemma-initialized vision-language backbone with a flow-matching action expert[7]. During project-specific fine-tuning, the pretrained vision-language backbone is frozen while the action expert and associated projection layers are optimized on the collected demonstrations. The deployed policy receives a natural-language task instruction, three camera observations, and a 16-dimensional dual-arm state, and produces continuous action chunks for robot execution. Training and deployment cover three bounded task families: ring placement, plug-and-socket insertion, and RAM manipulation. These tasks provide controlled test cases for object localization, grasping, placement, insertion, and language-conditioned task selection.

The scope of this thesis is a laboratory proof of concept rather than a production-ready assembly cell. Experiments are conducted with fixed robot hardware, bounded workspaces, defined initial conditions, and task-specific fixtures. The engineering targets—including task success, component detection, pose error, failure detection, and recovery—serve as acceptance criteria for the validation procedure and are not treated as achieved results unless supported by physical trials. The project does not claim unrestricted open-world generalization, autonomous factory deployment, or guaranteed safe recovery from every failure mode. Instead, it investigates whether a complete bimanual data and control pipeline can support reproducible, language-conditioned manipulation across a small but representative set of 3C assembly tasks.

1.3 Main contributions

The main contributions of this thesis are as follows:

1. An integrated dual-FR3 manipulation platform is established with GELLO teleoperation, synchronized three-camera sensing, robot-state and action recording, and task-compatible end effectors and fixtures.
2. A reproducible learning pipeline is developed to convert teleoperated demonstrations into datasets for ACT, Diffusion Policy, and $\pi_{0.5}$ training and real-robot deployment.
3. A pretrained VLA is adapted to the project workcell by fine-tuning its action-generation components, enabling one deployed policy to execute multiple bounded manipulation tasks from natural-language instructions.
4. A traceable engineering process links customer requirements and quality function deployment to concept selection, implementation decisions, and quantitative validation criteria.

1.4 Organization of this thesis

The remainder of this thesis is organized as follows. Chapter 5 translates customer requirements into measurable engineering specifications and presents the quality function deployment. Chapter 6 describes functional decomposition, concept generation, and the selection of the major hardware, sensing, teleoperation, and policy modules. Chapter 7 details the final system design, data and control architecture, model-training procedure, and implementation plan. Chapter 8 presents the physical validation methods and experimental results. Chapter 9 discusses the strengths and limitations of the system, summarizes the conclusions, and identifies directions for future work.

Chapter 2 Introduction

2.1 Background and motivation

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Chapter 5 Design specification

5.1 Customer requirements

Through discussions with the project sponsor and analysis of the target application, eight customer requirements were identified and assigned importance weights reflecting the sponsor's priorities. The requirements capture both functional goals and non-functional concerns, including robustness, safety, generalizability, cost, and reliability.

Table 5–1 Customer requirements and importance weights

No.	Customer Requirement	Weight
CR1	Complete assembly task successfully	0.18
CR2	High assembly precision	0.16
CR3	Robust to environment and disturbances	0.15
CR4	Safe and collision-free operation	0.14
CR5	Fast task execution	0.12
CR6	Easy to use and generalize	0.10
CR7	Cost-effective system	0.08
CR8	System reliable and recovery capable	0.07

The customer requirements were translated into engineering specifications through the QFD process rather than by assigning targets independently. For each customer requirement i and engineering specification j, the team assigned a relationship value $r_{ij} \in \{0, 1, 3, 9\}$ (none, weak, moderate, or strong) and calculated the unnormalized engineering priority as

$$I_j = \sum_i w_i r_{ij}, \tag{5–1}$$

where w_i is the customer-importance weight in Table 5–1. The priorities were then normalized to identify the technical characteristics that most directly protect the high-weight needs. CR1 and CR5 primarily drive ES1 and ES2; CR2 drives ES3 and ES4; CR3, CR4, and CR8 drive ES5 and ES6; and CR6–CR7 constrain the selected implementation to a usable, feasible, and maintainable route. This translation produces an auditable chain from the sponsor's language to acceptance tests and explains why the later concept search is organized around platform, teleoperation, data collection, gripper, and policy modules.

5.2 Engineering specifications

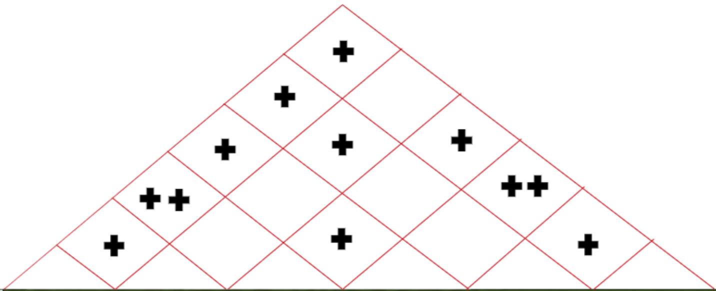
The customer requirements were converted into six measurable engineering specifications. The targets in Table 5–2 are the final acceptance thresholds for the present proof-of-concept, rather than performance claims. Each rate is calculated over independently initialized physical trials of the relevant task. A trial is counted once only; an unsuccessful recovery is not reclassified as an independent execution attempt. This convention prevents recovery behavior from artificially inflating the reported task-execution success rate.

Table 5–2 Final engineering specifications and acceptance thresholds

No.	Engineering Specification	Target	Operational Definition
ES1	Planning success rate	$> 90\%$	Fraction of trials in which the selected task policy and initialized scene form a valid executable plan.
ES2	Task execution success rate	$> 70\%$	Fraction of trials in which the prescribed assembly operation reaches its task-specific completion condition.
ES3	Component detection accuracy	$> 90\%$	Fraction of evaluated observations in which the required component is correctly detected for the active task.
ES4	Pose estimation error	$< 20\,\mathrm{mm}$	Euclidean position error between the estimated and reference component pose.
ES5	Failure detection rate	$> 90\%$	Fraction of induced or naturally occurring execution failures that are identified before further unsafe action is issued.
ES6	Recovery rate	$> 70\%$	Fraction of detected failures for which the defined recovery procedure restores a valid task state.

5.3 Quality function deployment

Figure 5–1 records the resulting House of Quality. Its 9–3–1 relationship scale makes the engineering priorities interpretable rather than decorative: task completion, component localization, pose estimation, and failure handling carry influence across multiple high-weight customer requirements. The roof is used as a design-control device. For example, additional visual coverage can strengthen ES3 and ES4, but it increases calibration, synchronization, and data-management burden; similarly, recovery logic can improve ES5–ES6 while adding controller integration work. These QFD trade-offs define the subsequent concept space. The platform and gripper must enable precise contact motion for ES1–ES2 and ES4; the tele-operator and data pipeline must produce usable demonstrations and observable scenes; and the policy must convert those observations into safe executable actions while leaving explicit hooks for failure monitoring and recovery.



Customer Requirements	Weight (Importance)	Engineering Specifications (ES)					
		1. Task Execution Success Rate (%)	2. Pose Error (mm)	3. Failure Detection Rate (%)	4. Recovery Success Rate (%)	5. Component Detection Accuracy (%)	6. Planning Success Rate (%)
Complete assembly task successfully	0.18	9	3	3	3	3	9
High assembly precision	0.16	3	9	1	1	9	3
Robust to environment and disturbances	0.15	3	3	9	9	3	1
Safe operation	0.14	3	1	3	3	1	1
Fast task execution	0.12	3	1	1	1	1	3
Easy to use and generalize	0.10	1	3	3	3	3	9
Cost-effective system	0.08	1	1	1	1	1	1
System reliable and recovery capable	0.07	3	1	9	9	1	3
Absolute Importance (Sum)		372	314	360	360	314	394
Relative Importance (%)		17.6	14.9	17.0	17.0	14.9	18.6
Rank (1 = Highest)		2	5	3	3	5	1
Benchmark		85%	3.2 mm	85%	75%	92%	88%
Target		>70%	<20 mm	>90%	>70%	>90%	>90%

Illustration 5–1 QFD House of Quality for the VLA bimanual manipulation system.

Chapter 6 Concept generation and selection

6.1 Concept generation

6.1.1 Functional decomposition

The target assembly system was first decomposed into solution-neutral functions: establish the workspace state; create a stable robot–operator interface for demonstrations; record synchronized visual, proprioceptive, and action data; grasp and constrain the component during contact; map the current observation to coordinated motion; and detect or recover from an invalid execution state. The decomposition converts the QFD priorities into five design modules. The robot platform provides repeatable bimanual motion for ES1, ES2, and ES4; the teleoperation module enables usable demonstrations; the data-collection module supports ES3–ES4 and learning reproducibility; the gripper realizes component contact; and the policy maps observations and state to actions. Failure monitoring and recovery remain cross-cutting functions because they act across the collection and deployment loop.

6.1.2 Structured brainstorming

The team used structured brainstorming to generate alternatives for each module before choosing a configuration. Candidate ideas were retained when they addressed at least one QFD-linked specification and did not violate laboratory safety, hardware access, or the project schedule. The session considered the available FR3 ecosystem alongside Flexiv and ALOHA-style platforms; direct joint-master, XR, and hand-held teleoperation; progressively richer observation schemas; compliant and rigid end-effectors; and policy families ranging from Diffusion Policy to ACT and future VLA models. This procedure deliberately separates the choice of a data-collection interface from the choice of a learning algorithm, since neither can be justified independently of the sensing and robot-control path.

The generated alternatives were also informed by established research directions rather than by ad hoc feature lists. Robomimic and MimicGen motivate treating demonstration quality and dataset construction as system-level design variables[8-9]; RVT, 3D Diffusion Policy, and R3M respectively motivate the retained alternatives for multi-view geometry, 3D visuomotor control, and reusable robot visual representations[10-12]. These references define the external concept space, while the selected configuration remains constrained by the laboratory hardware, the QFD priorities, and the project schedule.

6.1.3 Morphological chart and generated module concepts

Table 6–1 records the resulting concept space. A row represents one functional module, and a column entry represents one candidate realization. The bold entry in each row is the selected module concept; the table is not a claim that the unselected entries were all built or experimentally benchmarked.

Table 6–1 Morphological chart for the five functional modules

Concept	Candidate A	Candidate B	Candidate C
C1: Platform	**Dual Franka FR3**	Dual Flexiv platform	ALOHA-style bimanual platform
C2: Teleoperator	**GELLO joint-master**	PICO 4 Ultra XR	UMI hand-held
C3: Data collection	**Two wrist + one top camera + robot state/action**	One wrist + one top camera + robot state/action	Robot state/action only
C4: Gripper	Franka parallel gripper	3D-printed hard gripper	**3D-printed soft gripper**
C5: Policy	**ACT in LeRobot**	π-style VLA policy	Diffusion policy

The morphological chart produces five leading module concepts, denoted C1–C5. C1 is a dual-FR3 platform; C2 is GELLO-based teleoperation; C3 is a synchronized three-camera, robot-state, and action pipeline; C4 is a 3D-printed soft gripper; and C5 is an ACT policy integrated through LeRobot. Together they form one integrated system concept, rather than five competing end-to-end robot systems. Their alternatives are retained because each exposes a real engineering trade-off: platform precision against integration cost, operator intuitiveness against mapping complexity, observation coverage against synchronization burden, compliance against geometric repeatability, and policy capability against data and deployment maturity. Detailed descriptions of the unselected alternatives are provided in the appendix.

6.2 Concept selection

6.2.1 Selection method and QFD-derived criteria

Selection was conducted in two stages. First, an absolute screen removed a candidate if it could not be made safe in the laboratory, could not interface with the selected robot and data path, or could not be obtained and integrated within the project schedule. Second, the surviving alternatives were scored on a five-point ordinal scale, where 1 denotes a weak fit and 5 denotes a strong fit. Ratings were assigned from the QFD relationship to ES1–ES6, laboratory availability and interface knowledge, the existing collection workflow, and the cited primary or official sources; no rating is presented as an unreported benchmark result. Consistent with the functional decomposition, the "top five concepts" are the five selected module concepts C1–C5. Their combination forms the single system concept carried into the final design. For candidate a in module m, the decision score is

$$S_{m,a} = \sum_{k=1}^{4} w_k r_{m,a,k}, \tag{6–1}$$

where $r_{m,a,k}$ is the rating for criterion k and w_k is its QFD-derived weight. Scores are used only to compare alternatives within the same module; they are recorded design judgments, not measured task-performance results.

Table 6–2 QFD-derived criteria for module selection

No.	Criterion	Weight	QFD Link
SC1	Contribution to task performance and precision	0.30	ES1, ES2, ES3, ES4
SC2	Data, control, and software integration compatibility	0.25	ES1, ES2, ES3
SC3	Safety, robustness, and recovery support	0.20	ES5, ES6; CR3–CR4
SC4	Availability, manufacturability, and schedule feasibility	0.25	CR6–CR7 and project plan

6.2.2 Scoring of the five selected module concepts

Table 6–3 provides the detailed selection record for the five leading module concepts. The same criterion definitions and weights are applied in every module so that the scoring logic remains traceable to the QFD; however, the totals are interpreted only within a module because a robot platform, a gripper, and a policy make different contributions to the integrated system.

Table 6–3 Module-level concept selection matrix (1 = weak fit; 5 = strong fit)

Module	Candidate	SC1	SC2	SC3	SC4	Weighted score
C1 Platform	Dual Franka FR3	5	5	4	3	**4.30**
	Dual Flexiv	4	3	4	2	3.30
	ALOHA-style bimanual	3	4	3	5	3.85
C2 Teleoperator	GELLO joint-master	4	5	4	5	**4.50**
	PICO 4 Ultra XR	3	3	3	4	3.25
	UMI hand-held	3	3	3	4	3.25
C3 Data collection	Two wrist + top camera + state/action	5	5	4	3	**4.35**
	One wrist + top camera + state/action	4	4	4	5	4.20
	Robot state/action only	2	3	3	5	3.25
C4 Gripper	3D-printed soft gripper	4	4	4	5	**4.20**
	Franka parallel gripper	3	5	4	3	3.70
	3D-printed hard gripper	4	5	3	4	4.05
C5 Policy	ACT in LeRobot	4	5	4	5	**4.50**
	Diffusion Policy	4	3	4	4	3.75
	π-style VLA policy (future route)	5	2	3	1	2.75

6.2.3 Comparative evaluation and selected concept

C1: Dual Franka FR3 platform. The dual Franka FR3 platform was selected as the mechanical foundation of the final system because it provides repeatable 7-DoF control, a direct connection to the available laboratory control ecosystem, and a credible path toward coordinated bimanual manipulation. Its principal disadvantage is the substantial integration effort required for workspace coordination, collision avoidance, commissioning, and bilateral control. Nevertheless, these costs are justified by the project objective: lower-cost alternatives may reduce initial hardware burden, but they would introduce a different control ecosystem or provide a weaker fit to the required assembly precision.

C2: GELLO teleoperation. GELLO was selected as the planned demonstration-collection interface because it offers a direct and intuitive connection between operator motion and the robot-state/action records required by the learning pipeline[13]. Despite the need for reliable leader–follower mapping and calibration, GELLO remains preferable to immersive or hand-held alternatives for the present system because it provides the most direct route from tele-operated demonstrations to the joint-space data representation used by the FR3 and LeRobot workflow.

C3: Three-camera data pipeline. The selected sensing configuration uses one Intel RealSense D435 overhead camera and two Intel RealSense D405 wrist-mounted cameras, together with synchronized robot state, gripper state, and action records. The D435 provides a stable global view of the shared workspace, while the two D405 cameras preserve local visual information near the end-effectors and contact regions. This complementary arrangement reduces visual blind spots and directly supports the component-detection and pose-estimation requirements in ES3–ES4, at the cost of increased synchronization and data-management effort.

C4: 3D-printed soft gripper. The selected end-effector is a 3D-printed soft gripper derived from the open-source gripper design in the referenced UMI material; the project adopts only the gripper-finger component rather than reproducing the complete UMI hardware system[14]. Compliant contact and rapid geometry iteration are valuable for handling assembly components with small geometric variation, although deformation may reduce pose repeatability compared with a rigid locating gripper.

C5: ACT policy in LeRobot. ACT was selected as the current policy because it is integrated with the LeRobot workflow and supports task-specific training and checkpoint deployment from synchronized visual observations and robot-state inputs[1,15]. Action chunking provides a practical temporal-control mechanism for the present imitation-learning pipeline. Deployment remains task-specific and operator-selected: the policy does not yet infer tasks from natural-language input or demonstrate unified multi-task control. This limitation is accepted because ACT provides a deployable and traceable baseline for the current project stage, whereas Diffusion Policy remains an earlier baseline and π-style VLA models remain a future extension.

The selected architecture is the combination of the highest-scoring surviving candidate in each module. C1 and C4 establish the physical capability for controlled contact; C2 and C3 make demonstrations and visual evidence available to the learner; and C5 provides the deployable policy path. This combined decision covers the full ES1–ES6 chain while retaining a feasible implementation route for subsequent final design, manufacturing, and validation.

Chapter 7 Final design and implementation plan

7.1 Final design

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7.2 Implementation plan

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Chapter 8 Validation results

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Chapter 9 Discussion and conclusions

9.1 Discussion

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9.2 Conclusions

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9.3 Future works

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Engineering changes notice Appendix 1

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Bill of materials Appendix 2

Table 2–1 Bill of materials

Quantity	Part Description	Purchased From	Part Number	Price (each)	Notes
…	…	…	…	…	…

Research Projects and Publications during Undergraduate Period

[1] ……

Acknowledgements

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A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

HCCI (Homogenous Charge Compression Ignition) combustion has advantages in terms of efficiency and reduced emission. HCCI combustion can not only ensure both the high economic and dynamic quality of the engine, but also efficiently reduce the NOx and smoke emission. Moreover, one of the remarkable characteristics of HCCI combustion is that the ignition and combustion process are controlled by the chemical kinetics, so the HCCI ignition time can vary significantly with the changes of engine configuration parameters and operating conditions. ⋯⋯